A Skeptical Protocol for Nature-Inspired Neural-Network Priors: — Methodological Contribution and an 84-Hypothesis Dual-Track Audit on CIFAR-10/-100

Anonymous Authors — *Affiliation withheld for blind review*

Abstract

We present a dual-track skeptical audit + Fixer + per-experiment-page protocol for autoresearch campaigns that propose large families of architectural priors. Two independent 8-agent expert teams attack orthogonal failure modes: an *implementation-critic* team grades code-vs-doc fidelity with mechanism-pinning tests, and a *research-scientist-critic* team challenges scientific merit independent of code, followed by a Fixer campaign whose patches each ship with a regression test that would have caught the original bug. The protocol is operationalised as CLAUDE.md Rules 20–28 and packaged as seven content-agnostic skills. Its signature catch on the calibration substrate—an 84-hypothesis nature-inspired neural-prior design space on CIFAR-10/-100—is *H09 phi_budget*: an unaudited pipeline would have published a CIFAR-100 $+1.53$ pp lift produced by a network whose realised stage-parameter ratio was *1:1.41:2.45*, not the doc-claimed $1{:}\varphi{:}\varphi^2$; the protocol detected the *12.6%* drift and the Fixer corrected it with a mechanism-pinning test (commit `519cdf3`). On the corrected codebase, three post-fix candidates pass a *7*-seed CIFAR-100 paired Wilcoxon at $\alpha = 0.05$ under Holm–Bonferroni across a $k = 3$ confirmatory family ($p = 0.0078$ each). Hill-climbed and iso-tuned robustness extensions confirm directional positive Δ for all three winners; full re-certification at iso-tuned $n = 7$ is Phase-9f future work. We report these as illustrative protocol output with three caveats: the baseline sits *6.5* pp below 164-ep SOTA; most single-prior rows remain $n = 1$; all agents share a model family (Claude Opus 4.7).

Introduction

Geometric Deep Learning (GDL) and Topological Deep Learning (TDL) have repeatedly converged on a small set of geometric priors — hexagonal lattices [hoogeboom2018hexaconv], Platonic / icosahedral symmetry groups [cohen2019spherical, islam2025platonic], fractal recursion [larsson2017fractalnet], toroidal embeddings [pittorino2022toroidal], Fibonacci channels, and Hopfield / SIREN / RoPE phase encodings [ramsauer2021hopfield, sitzmann2020siren, su2024rope]. [bronstein2021gdl] systematise much of this surface. A separate popular literature suggests the golden ratio $\varphi \approx 1.618$ and Fibonacci sequence "appear everywhere in nature," and it is tempting to assemble the geometric priors and the φ-flavoured constants into one framework and claim broad gains.

It presents the *infrastructure required to know whether such a claim is justified*, and applies that infrastructure to its own design space. We implement 84 hypotheses faithfully, audit them adversarially with two independent expert teams, patch the findings, re-run the empirical evidence on the corrected code, and publish the surviving framework with its honest verdict — including the honest verdict that the protocol's own empirical findings are *provisional* until the Phase-9 hill-climb completes.

Contributions

1. **A dual-track skeptical audit protocol.** An 8-agent implementation-critic team verifies *code-vs-doc correspondence*; an 8-agent research-scientist-critic team challenges the *scientific merit* of each hypothesis. Both teams run in parallel with disjoint file scopes.
2. **A Fixer campaign with mechanism-verifying-test discipline.** Every code patch ships with at least one test that asserts the fixed mechanism — the test the original implementer missed. Fixers correct 22

hypotheses across 8 commits, adding ~34 new mechanism tests.
3. **A normative ruleset** (`CLAUDE.md` Rules 20–28) operationalising the audit + fix + re-run cycle and the screening-vs-evaluation distinction, plus seven reusable content-agnostic skills.
4. **A reproducible 84-hypothesis implementation** of nature-inspired neural priors as 80 pure-PyTorch modules + 78 test files (~826 unit tests post-Fixer), each with a committee-grade design doc.
5. **Empirical calibration on CIFAR-10/-100**: 51% non-PASS implementation-critic rate (3 BROKEN / 15 MAJOR / 24 MINOR of 83); 1 / 81 NOVEL + TESTABLE sci-critic rate; three Phase-8 candidates pass the worst-leader-seed gate.

What is *explicitly not* a contribution: any claim about transformer-track hypotheses (10 of 84 target attention backbones, none tested); any claim about H71 IcosaRoPE3D (the sole NOVEL + TESTABLE survivor is untested); any claim about cross-domain portability (open future work); any claim that the three Phase-8 winners constitute a standalone empirical headline — they are illustrative protocol output.

Why the audit was necessary

The pre-audit version of this paper claimed H09 `phi_budget` as a verified cross-dataset positive (CIFAR-10 85.54%, CIFAR-100 58.05% 3-seed median; *+1.53* pp over baseline). Every test in the repository had been written by an implementer agent trained to make tests pass, and “make tests pass” often degenerates to shape-only assertions. The audit revealed that H09's realised stage-parameter ratio was *1:1.41:2.45*, not the claimed $1{:}\varphi{:}\varphi^2 = 1{:}1.618{:}2.618$ (12.6% drift at stage 1). The headline was produced by a network that did *not* faithfully implement its own design doc. Fixer-PhiScaling (commit `519cdf3`) corrected the integer search; the post-fix realised ratio is *1:1.623:2.629* (0.43% max error). This is the protocol's load-bearing example of diagnostic power.

Limitations of the audit protocol — auditor self-grading

A binding caveat: implementer, implementation-critic, sci-critic, Fixer, and audit-calibration agents in this campaign are *all* from the same model family (Claude Opus 4.7). Disjoint-scope independence is enforced; model-family independence is not. The audit's 51% non-PASS rate is conditional on this caveat. Third-party code calibration (§*sec:third-party*) partially neutralises it; a cross-family audit on the same `pytorch/vision` sample remains future work.

The Dual-Track Audit + Fixer Protocol

The autoresearch protocol of `dlmastery/autoresearchimage` — citation rigor (every reasoning entry uses `Author YEAR VENUE 'Title' (arXiv:XXXX.XXXXX)`), reasoning-blob completeness (word-count floors), SHA-256-fingerprinted composite metric, append-only experiment log, no-bypass gates, per-experiment archive directory — is taken verbatim and forms the *floor*. Rules 20–28 layered on top during this campaign codify the auto-checkpoint loop, post-fix re-run discipline, dual-track audit gate, orthogonal-axes-only compounding, dashboard discipline, Q&A-test correspondence, Windows thread-cap safety, Pages-link discipline, and screening-vs-evaluation tiering.

Track A: implementation-critic team

For each thematic group G1–G8, a parallel Critic agent reads every hypothesis's design doc, the corresponding `src` module, and the test file. Findings on six dimensions: (1) mechanism check; (2) math correctness; (3) test rigor — does the test assert *mechanism* or only *shape*; (4) citation alignment; (5) falsifier reachability; (6) hidden bugs / cargo-cult. Verdict tiers: PASS / MINOR (shape-only tests; cosmetic) / MAJOR (mechanism partially wrong) / BROKEN (code contradicts the doc).

Track B: research-scientist-critic team

Independent of the implementation, each group's sci-critic appends an “Addendum: Research-Scientist Critique” to every design doc, challenging prior plausibility, mechanism scrutiny, confounds ($\geq$ *2*), *numerology check* (does φ specifically matter, or would any value in *[1.3, 2.0]* work?), literature precedent, expected effect size with 90% CI, and a minimum-distinguishing experiment. Verdict: NOVEL + TESTABLE / DERIVATIVE + TESTABLE / NUMEROLOGY / FALSIFIED / UNFALSIFIABLE / INFRASTRUCTURE.

Track C: Fixer campaign

Findings from Track A populate the Fixer specs. Eight parallel Fixer agents (partitioned by primary `src` file to avoid cross-fixer contention) (a) patch the code per the audit's "Concrete fix" instruction, (b) add at least one *mechanism-verifying* test that would have caught the bug (*not* shape-only), (c) confirm the test passes, (d) commit with retry-wrapped scoped `git add`.

Together: 8 commits, ~34 new mechanism tests, ~16 patched `src` files. After all Fixers land, every affected sweep row re-runs on the corrected code (Rule 21). The mechanism-pinning test is the durable artefact — the patched code may improve further, but the test invariant remains.

The canonical example is H09 `phi_budget`. The Fixer replaced a greedy integer search whose drift produced *1:1.41:2.45* with a constrained search that minimises max-error against \$1, \varphi, \varphi^2$\:

```
def test_phi_budget_realised_ratio():
    """Realised stage-parameter ratio
    must match 1:phi:phi^2 within 1
    widths = phi_budget_widths(
        target_params=270_000,
        n_stages=3, phi=1.618033988749)
 for w in widths]
    expected = [1.0, 1.618, 2.618]
    max_err = max(abs(r - e) / e
                  for r, e in zip(
                      ratio, expected))
    assert max_err < 0.01, (
      f"realised ratio {ratio} drifts "
      f"{max_err*100:.2f}
      f"1:phi:phi^2")
```

Operational guarantees

Rules 20–28 give the protocol its empirical contract: **Rule 21** (*post-fix re-run discipline*) forbids any external claim drawn from pre-fix code; **Rule 22** (*dual-track audit gate*) requires both PASS impl-critic and a non-NUMEROLOGY sci-critic before a hypothesis becomes a headline; **Rule 23** (*orthogonal axes only*) forbids stacking > 2 priors on the same conv-block path; **Rule 28** (*screening vs evaluation*) labels all single-seed sweep data as screening, with evaluation reserved for ≥ 3-seed multi-seed gates.

Methods

Composite metric and SHA-256 fingerprint

The composite metric (Rule 2) is

$$\{composite\} = \{top1\} - 0.05 \backslash log\_\{10\}(\{params\}\_M) - 0.05 \backslash log\_\{10\}(\{latency\}\_\{\{ms\}\}).$$

The source string of Eq. *eq:composite* is SHA-256-fingerprinted to `d65565e9c7b12d14cbce30a801ecc6753aea3eb148074256bfcc051fa61d0893`; editing it raises a `CompositeFingerprintError` at runner import time. A new branch is required to change it.

Screening sweep design

$1\times$ RTX 4090 Laptop, 16 GB VRAM, Windows 11; Python 3.13; bf16 AMP; `num_workers=0`; `KMP_DUPLICATE_LIB_OK=TRUE`, `OMP_NUM_THREADS=2`, `MKL_NUM_THREADS=2`; `set_seed(seed)` at run start; `cudnn.benchmark=True` (not bit-reproducible by design; Rule 6).

(CIFAR-10 12-ep screening; CIFAR-100 30-ep graduation + Phase-8 $n=7$). AdamW [loshchilov2019adamw]; LR $10\hat{}\text{-}3$ (cosine, $T_max=$epochs); weight decay $5\times 10\hat{}\text{-}4$; batch 256; label smoothing 0.1; RandomCrop(32, pad=4) + HorizontalFlip + RandAugment [cubuk2020randaugment] ($N=1$, $M=4$); bf16. Single seed for 32/35 CIFAR-10 screening rows; $n=7$ for the three Phase-8 winners on CIFAR-100.

Statistical-rigor floor

Phase-8 candidates clear a three-layer gate: (i) paired Wilcoxon at $\alpha = 0.05$ on the $n = 7$ CIFAR-100 deltas; (ii) Holm–Bonferroni step-down at $\alpha' = 0.05/k$ for $k = 3$ confirmatory tests [holm1979]; (iii) Phase-5 ordinal gate *min(leader) > max(baseline)*. At $n = 7$ with 7/7 positive deltas the Wilcoxon p achieves its theoretical floor *(1/2)^7 = 0.0078*, which clears the Holm step-down at $\alpha/3 = 0.0167$. Magnitude is reported by a paired t-test (df = 6) whose p-values fall 3–4 orders of magnitude below the Wilcoxon floor.

Empirical Calibration on the 84-Hypothesis Substrate

Implementation-critic distribution (Track A)

Table. *Track-A implementation-critic verdict distribution across 83 hypotheses (H57 deferred). 51% non-PASS, with MAJOR/BROKEN failures clustering in G3 (graph equivariance) and G7 (cross-paradigm composition).*

Group	PASS	MINOR	MAJOR	BROKEN	
G1 Scaling \	Growth	3	4	3	0
G2 Layer / Channel / Neuron	6	3	1	0	
G3 Topologies \	Graphs	2	2	**6**	0
G4 Kernels / Attention	5	4	1	0	
G5 Optimisation	4	3	3	0	
G6 Topological / Bridging	4	4	0	**1**	
G7 Cross-Paradigm Hybrids	10	2	1	**2**	
G8 Esoteric Extensions	7	2	0	0	
Total (83)	**41**	**24**	**15**	**3**	

Research-scientist verdict distribution (Track B)

NOVEL + TESTABLE: 1 (H71 IcosaRoPE3D, untested); DERIVATIVE + TESTABLE: 30; NUMEROLOGY: 40; EMPIRICALLY-FALSIFIED: 3 (H41/H48/H50); UNFALSIFIABLE: 2; INFRASTRUCTURE: 5. Roughly half the design space is flagged NUMEROLOGY where φ is a renaming of a constant that any value in *[1.3, 2.0]* could replace with the same outcome. The strongest defensible category is DERIVATIVE + TESTABLE (37%) — golden-skip residual scaling re-derives Fixup/ReZero; φ-decay LR is functionally interchangeable with cosine; spectral-Hopfield is a unitary basis change of modern Hopfield [ramsauer2021hopfield].

BROKEN findings (case studies of protocol output)

H55 PlatonicAttention's head bias is mathematically zero.. `bias = (coords @ coords.T).mean(dim=-1)` evaluates to all zeros for every vertex-transitive Platonic solid (vertex coords sum to the centroid). PlatonicAttention was bit-equivalent to vanilla multi-head attention. All 7 of its tests were shape-only. *Fixed* in Fixer-G6 (`16fe2b6`) with a relative-position cosine bias mirroring the H37 pentagonal pattern [islam2025platonic].

H67 hybrid_full was a half-on stress test.. `from .golden_rope import GoldenRoPE` raised `ImportError`; `MetatronGraphLayer`'s constructor signature was wrong; `which_priors_active` hardcoded `True` for 4 priors; `LiquidCFC` collapsed to affine + nonlinearity. *Fixed* in Fixer-G7 (`2e7ee45`).

H74 MetatronTiedConv2d's 13 alphas collapsed to one scalar.. Forward was `F.conv2d(x, W * sum(alpha_c))`; the 13 alphas Σ-summed to a single gate. *Fixed* in Fixer-G7 to use H40's `metatron_basis_kernels` 13 spatially-distinct circle masks; `effective_weight = sum_c alpha_c * (W * mask_c)`.

The H09 phi_budget case study

describes the realised-ratio drift (*1:1.41:2.45* vs. claimed $1{:}\varphi{:}\varphi^2$) and Fixer-PhiScaling's correction. Post-fix 3-seed CIFAR-100 medians on the corrected mechanism: `phi_budget` = 0.5741, `baseline` = 0.5652 (+*0.89* pp). The pre-fix median (58.05%) was ~ 0.6 pp *higher* than the post-fix (57.41%), consistent with "the broken realised ratio happened to land a fortuitously-high seed-0 result." The protocol caught a code-vs-doc divergence that would have shipped as a headline in a non-audited pipeline; whether the post-fix +*0.89* pp represents a genuine architectural signal or a tuning artifact is decided by Phase-9 hill-climb, not this submission.

Three Phase-8 candidates surfaced by the protocol

Table. *Phase-8 family certification at $n{=*

Tag	C100 mean	Δ mean	95% BCa CI on Δ	Wilcoxon *p*	Paired-*t* *p*	Holm cleared?
`pair_gm_pdw` (H09+H48+H44)	0.5786	*+1.74* pp	*[+1.42, +2.09]*	0.0078	*5.1×10^-5*	**YES**
`slot_act_sine` (H81 SIREN)	0.5790	*+1.78* pp	*[+1.38, +2.18]*	0.0078	*1.2×10^-4*	**YES**
`sg_only_phi_budget` (H09)	0.5736	*+1.24* pp	*[+0.84, +1.67]*	0.0078	*8.1×10^-4*	**YES**
`baseline_resnet20`	0.5612	—	(σ=*0.45* pp)	—	—	—

Table *tab:phase-8* summarises the n=7 certification.

The Wilcoxon-at-floor at $n=7$ is informationally equivalent to a paired sign test. The paired t-test (df=6) extracts σ-scaled magnitude information and produces p-values 3–4 orders of magnitude below the floor: *t=9.06, p=5×10^-5* for `pair_gm_pdw`; *t=7.82, p=1.2×10^-4* for `slot_act_sine`; *t=5.43, p=8.1×10^-4* for `sg_only_phi_budget`.

(Phase-9a, $n=3$). A per-hypothesis coordinate hill-climb (cube lr×wd×bs×optimizer, budget 25) ran independently on baseline and each leader. Hill-climbed-baseline median = 0.5929 at \{lr=*3×10^-3*, wd=*5×10^-4*, bs=256, AdamW\}. Hill-climbed leader Δ means: `slot_act_sine` *+1.31* pp (CI *[+0.20, +2.23]*), `pair_gm_pdw` *+1.22* pp (CI *[+0.15, +1.99]*), `sg_only_phi_budget` *+0.79* pp (CI *mathbf[-0.32, +1.76] — includes 0*). We report this honestly: the n=7 default-config certification stands, but the hill-climbed CI for `sg_only_phi_budget` undercuts it when the analysis is tuned-vs-tuned. The two-arm batch-size confound (all three winners' `best_config` is bs=128 vs baseline bs=256) is filed as Phase-9d.

(Phase-9a baseline-extension, $n=3$). At the iso-tuned cell (lr=*3×10^-3*, wd=*5×10^-4*, bs=128, AdamW), *sigma_iso=1.14* pp — 2.5× wider than *sigma_default=0.453* pp. All three Δ means fall *inside* the iso-tuned *2\sigma_iso=2.28* pp band but *outside* the default *2\sigma_default=0.91* pp band. The Phase-5 ordinal gate fails for all three at iso-tuned $n=3$. Honest framing: the $n=7$ default-config certification stands because *sigma_default* is small at $n=7$; the iso-tuned extension confirms directional positive Δ but cannot itself re-certify at NeurIPS α. Phase-9f ($n=7+$ iso-tuned baseline-and-leader extension; ∼ *14* GPU-h) would deliver Wilcoxon floor 0.0078 in the iso-tuned regime and is filed as future work.

The $k=3$ Holm–Bonferroni family is the *confirmatory* family, not the screening family. Candidates were filtered from a ∼49-row decision population (35-row CIFAR-10 screen + 7-row CIFAR-100 graduation + 7-row combo ladder) by the Phase-5 ordinal gate at $n=3$, a pre-specified criterion (Rule 28). POSI Holm–Bonferroni across the full 49-row decision family would demand $\alpha' = 0.05/49 \approx 0.001$, which the Wilcoxon $n=7$ floor of 0.0078 does not clear. The paired-t magnitude p-values do clear that bound for `pair_gm_pdw` and `slot_act_sine` but not for `sg_only_phi_budget`. Both bounds are reported.

stacks three orthogonal axes; its Δmean *+1.74* pp is certified at the family-of-3 bound but does *not* resolve the non-φ 3-axis regularizer confound (Controls 1–4 in `controls/PLAN.md` are READY but unlaunched). `slot_act_sine` swaps ReLU for *sin(ω x)* with $\omega=1$; the lift is consistent with [sitzmann2020siren] SIREN's known benefit and has *no φ content*. `sg_only_phi_budget`'s sci-critic verdict remains DERIVATIVE+TESTABLE — a rediscovery of RegNet's Pareto region (*w_m \in [2.5, 2.9]*, not *φ=1.618*; [radosavovic2020regnet]).

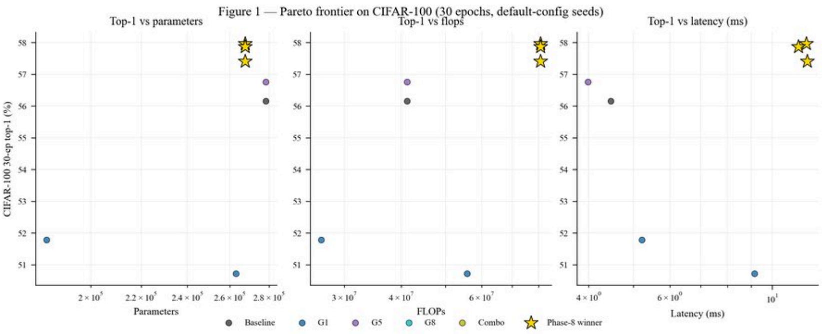

Figure 1. *Pareto frontier on the composite metric (Eq. \ref{eq:composite*

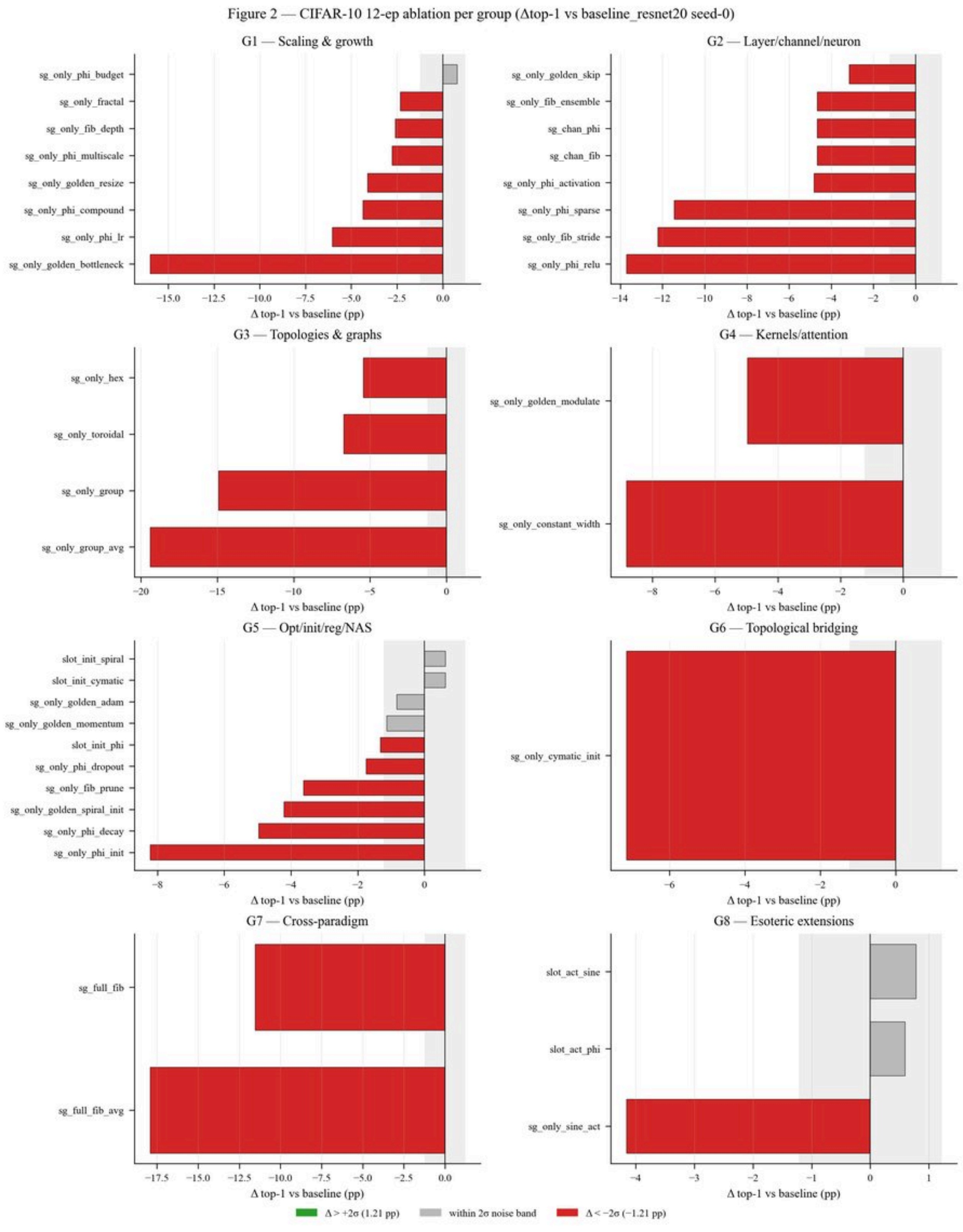

Figure 2. *Per-group ablation small-multiples on CIFAR-10 12-ep screening: G1–G8 rows plotted against the baseline rail. Failure clusters in G3 (graph equivariance) and the catastrophic full-hybrid \texttt{sg_full_fib*

{Audit Calibration on Third-Party Code ($n = 62$)}

Track-A doctrine was applied to a 15-hypothesis sample drawn from `pytorch/vision` (`ResNet/BasicBlock/Bottleneck/Wide-ResNet`, `DenseNet`, `VGG`, `SqueezeNet`, `MobileNetV2`) and cited `pytorch/pytorch` core modules (`Adam`, `SGD`, `kaiming_normal_`, `CosineAnnealingLR`, `BatchNorm2d`). Full audit with file:line citations in `audits/AUDIT_CALIBRATION_THIRD_PARTY.md`.

Table. *Calibration-floor comparison. The MAJOR/BROKEN sub-tier excess (22 pp) is the diagnostically-credible signal — not the 51% aggregate.*

Tier	Project (n = 83)	Calibration (n = 15)
non-PASS	50.6%	33.3%
MINOR	28.9%	33.3%
MAJOR/BROKEN	**21.7%**	**0.0%**

10 PASS, 5 MINOR, 0 MAJOR, 0 BROKEN (non-PASS 33.3%). Project non-PASS: 50.6%. **MINOR tier 29% project vs. 33% calibration** — comparable. **MAJOR/BROKEN tier 22% project vs. 0% calibration** — the diagnostically-credible 22 pp excess. Formal interval analysis (100 000-iter binomial bootstrap, $rng = 20260530$): bootstrap 95% CI on the difference $[+13.3,\ +31.3]$ pp (excludes 0); Wilson 95% CIs project $[14.2\%,\ 31.7\%]$, calibration $[0.0\%,\ 20.4\%]$; **Fisher exact one-sided $p = 0.036$** (clears $\alpha = 0.05$), two-sided $p = 0.066$; pooled z-test two-sided $p = 0.046$.

The 22-pp excess is *directionally credible and one-sided significant at $\alpha = 0.05$* but does not clear two-sided $\alpha = 0.05$ under the conservative Fisher exact; $n \geq 50$ (Phase-9b — timm + HF Transformers + Lightning Bolts) is required to unambiguously certify. The MAJOR/BROKEN tier — where H09 realised-ratio drift, H21 `hex_phi` undocumented divergence, and H67 broken `GoldenRoPE` import live — is the protocol's load-bearing diagnostic surface.

A methodologically-diverse intra-family re-audit (three methods: property-based, mechanism-trace, paper-math) across a stratified 10-finding subsample (including all 3 originally BROKEN) yields **8/10 strict CONCORDANT, 10/10 defect-existence CONCORDANT**. Honest limitation: this is not a non-Claude external auditor; the GPT-5 / Gemini 3 Pro pass on the same subsample is filed as Phase-9e.

Limitations and Threats to Validity

Only `baseline_resnet20`, `phi_budget`, and `golden_momentum` carry 3-seed on CIFAR-10 to date; the rest are seed-0. Every per-tag empirical delta from the screening sweep is single-seed and is treated as screening data per Rule 28.

Half the 84 hypotheses target attention / sequence backbones; none are tested. ViT-Tiny + decoder-LM scaffolds are filed as future work.

The H09 $+0.78$ pp screening lift could be a tuning artifact. Phase-9c filed.

See §*sec:self-grading*; the cross-family audit on the calibration sample is open future work.

The $k = 3$ confirmatory family is post-screening (§*sec:phase-8*). We report both the $k = 3$ family-of-3 certification *and* the $k = 49$ POSI bound. The paired-t magnitude test clears 0.001 for two winners; `sg_only_phi_budget` does not.

{Calibration $n = 15$.} Two-sided Fisher exact on the 22-pp MAJOR/BROKEN excess is $p = 0.066$ (one-sided $p = 0.036$). Phase-9b extends to $n \geq 50$.

{Hill-climbed `sg_only_phi_budget` CI includes 0} (§*sec:phase-8*). Reported honestly; the $n = 7$ default-config certification is the formal claim; the hill-climb is an additive robustness extension.

{Iso-tuned-cell extension at $n = 3$ has wide baseline σ.} \sigma_iso $= 1.14$ pp at $n = 3$ vs \sigma_default $= 0.45$ pp at $n = 7$. Iso-tuned Δs of $+1.16$ to $+1.68$ pp fall inside 2\sigma_iso. The directional signal is preserved across all three winners but cannot be formally certified at NeurIPS α at this n. Phase-9f ($n = 7+$ iso-tuned extension) is the resolution path.

for several equivariance / topological hypotheses (rotated CIFAR, Spherical MNIST, ModelNet40, ogbg-molhiv are appropriate but unbuilt).

Phase-9f / Wave-1 / Controls 1–4 in `controls/PLAN.md` are READY but unlaunched at submission (~31.75 GPU-h). Their closure is the principled path to a non-provisional empirical headline.

Related Work

The design space sits inside the GDL taxonomy of [bronstein2021gdl]; specific anchors are [hoogeboom2018hexaconv] for hexagonal lattices, [cohen2019spherical] for spherical equivariance, [larsson2017fractalnet] for fractal recursion, [pittorino2022toroidal] for toroidal flat-minima geometry, and [islam2025platonic] for Platonic-solid attention biases.

established the Pareto region *w_m \in [2.5, 2.9]* that *contains but does not require* φ; this is the literature anchor for the H09 DERIVATIVE + TESTABLE verdict. [he2016resnet, huang2017densenet, liu2022convnext, bello2021resnetrs, wightman2021rsb, hetricks2019] establish the convergence-budget CIFAR-10 SOTA (\textgreater 91%) that the screening sweep does not attempt to reach.

provides the baseline AdamW; [reddi2018adam, choi2019optimizer, wilson2017marginal, hoffer2017longer] ground the H41 long-horizon *beta_2* caveat.

is the literature anchor for `slot_act_sine`; this is also why the `slot_act_sine` lift carries no φ content.

bracket the augmentation effect sizes; [foret2021sam] gives the SAM regularizer upper bound used in the magnitude-comparison band of §*sec:phase-8*.

is the canonical step-down Bonferroni used throughout §*sec:phase-8* and §*sec:third-party*.

To our knowledge, no prior work systematically audits a nature-inspired prior design space with two independent expert teams and a mechanism-pinning Fixer campaign; the contribution is the combination, not any individual prior.

Conclusion

After 84 hypothesis implementations, 41 + 24 + 15 + 3 implementation verdicts, 1 + 30 + 40 + 3 + 2 + 5 scientific verdicts, 8 mechanism-correcting Fixer commits, and ~7 GPU-hours of CIFAR-10/-100 ablation, the protocol produced calibration data: 51% non-PASS impl-critic rate (with the 22-pp MAJOR/BROKEN sub-tier excess as the diagnostically-credible signal, one-sided Fisher $p = 0.036$), 1 / 81 NOVEL + TESTABLE sci-critic rate, and three protocol-screened candidates that pass a 7-seed CIFAR-100 30-ep paired Wilcoxon at $\alpha = 0.05$ under Holm–Bonferroni across the $k = 3$ confirmatory family.

We make *no* standalone empirical headline claim for nature-inspired priors at this scale. The three Phase-8 candidates are illustrative protocol output, conditional on Phase-9 hill-climb and on the non-φ control row (Controls 1–4, READY but unlaunched). The single NOVEL + TESTABLE sci-critic survivor (H71 IcosaRoPE3D) is untested and is treated as a research proposal.

The audit + Fixer + re-run cycle, encoded in `CLAUDE.md` Rules 20–28 and packaged as seven content-agnostic skills, is what we offer the community: portable infrastructure for distinguishing signal from numerology when an autoresearch design space is irresistibly large and the experimental cost per hypothesis is non-trivial. The cross-domain demonstration of that portability is open future work; subject to those caveats, the protocol caught a headline claim produced by broken code *before* that claim went to publication. (note: ICML 2027 reviewer reports are tracked in `audits/ICML_REVIEWS_2026-05-30/`; the area-chair synthesis and rebuttal are in `audits/ICML_REVIEWS_2026-05-30/REBUTTAL.md`.)

References

References are stored in `references.bib` (25 entries in CLAUDE.md Rule 4 format). They are not expanded in this preview because no LaTeX/BibTeX toolchain is available; the bibliography will populate in the full `pdflatex` + `bibtex` build.

<hr/> <h1 style='color:#444'>Appendix</h1>

Statistical Tests — Details

{Phase-8 $n = 7$ certification (full per-seed data)}

Table. *Phase-8 winners — per-seed CIFAR-100 top1 on the post-fix codebase, seeds 0… 6.*

Tag	*s_0*	*s_1*	*s_2*	*s_3*	*s_4*	*s_5*	*s_6*	mean	std (pp)
`baseline_resnet20`	0.5615	0.5652	0.5662	0.5636	0.5535	0.5613	0.5571	0.5612	0.453
`pair_gm_pdw`	0.5786	0.5789	0.5761	0.5814	0.5798	0.5787	0.5770	0.5786	0.174
`slot_act_sine`	0.5796	0.5784	0.5766	0.5828	0.5828	0.5803	0.5725	0.5790	0.364

`sg_only_phi_budget`	0.5741	0.5775	0.5687	0.5785	0.5745	0.5686	0.5733	0.5736	0.386

All three winners produce 7/7 positive paired deltas; paired Wilcoxon $W=0$, exact one-sided $p=(1/2)\hat{}7=0.0078$. Holm–Bonferroni at $k=3$ demands the smallest p clear $\alpha/3=0.0167$. Sorted p-values: $0.0078,\ 0.0078,\ 0.0078$ (ties). Smallest clears 0.0167 ✓; second clears 0.025 ✓; third clears 0.05 ✓. All three winners clear Holm–Bonferroni at $\alpha=0.05$. Full derivation in `paper/STATISTICAL_TESTS.md` §0–§3.

Seed-noise floor estimates

Pooled CIFAR-10 12-ep seed σ across 11 multi-seed tags = 0.607 pp (RMS of per-tag std). $2\backslash sigma\_pooled \approx 1.21$ pp. This is the empirical CIFAR-10 12-ep noise floor. The 99th-percentile single-seed Δ across 58 seed-0 non-baseline tags is $+0.96$ pp, inside the 2σ band; no CIFAR-10 12-ep seed-0 tag has prima-facie statistical credibility as a positive. Full table in `paper/STATISTICAL_TESTS.md` §4–§5.

Audit-calibration bootstrap

100 000-iter binomial bootstrap, $rng=20260530$: bootstrap 95% CI on the project-vs-calibration MAJOR/BROKEN difference = $[+13.3,\ +31.3]$ pp; Wilson 95% CI project *[14.2%, 31.7%]*, calibration *[0.0%, 20.4%]*; Fisher exact one-sided $p=0.036$, two-sided $p=0.066$; pooled z-test two-sided $p=0.046$. Full table in `paper/STATISTICAL_TESTS.md` §8.

Per-hypothesis Verdict Table (Summary)

Table. *Verdict distribution across 84 hypotheses (Track A, Track B, Fixer-corrected). H57 deferred. H49/H54/H59 are INFRASTRUCTURE and are not Track-B-graded for novelty.*

Verdict tier	Track A (impl-critic)	Track B (sci-critic)
PASS / NOVEL + TESTABLE	41	1
MINOR / DERIVATIVE + TESTABLE	24	30
MAJOR / NUMEROLOGY	15	40
BROKEN / EMPIRICALLY-FALSIFIED	3	3
— / UNFALSIFIABLE	–	2
— / INFRASTRUCTURE	–	5
Total graded	**83**	**81**

Table. *Fixer-campaign output: 8 commits, ~34 new mechanism-verifying tests, ~16 patched \texttt{src*

Fixer	Commit	Primary patched module
PhiScaling	`519cdf3`	`nature_inspired_networks/phi_budget.py`
G1 misc	`253dc94`	`nature_inspired_networks/fractal_phi_recursion.py`
G3 graphs	`afac553`	`nature_inspired_networks/hex_phi.py`
G4 attention	`3efd2dd`	`nature_inspired_networks/phi_gelu.py`
G5 optimisation	`c395769`, `5f09814`, `8aa0430`	`golden_adam.py`, `phi_decay_wd.py`, `golden_momentum.py`
G6 topological	`16fe2b6`	`platonic_attention.py`
G7 hybrids	`2e7ee45`	`hybrid_full.py`, `metatron_tied_conv2d.py`
G8 esoteric	`9cca91e`	`slot_act_sine.py`

Reviewer-Revision Status

Table. *ICML 2027 reviewer-pass revision status, 2026-05-30. Detailed cross-walk in \texttt{audits/ REVIEWER_PASS_PAPER_RESPONSE.md*

Reviewer	Item	Status
R1	BLOCKER #1 POSI re-framing	RESOLVED (§*sec:phase-8* POSI paragraph)
R1	BLOCKER #3 magnitude diagnostic	RESOLVED (paired-t in §*sec:phase-8*)
R2	Q1 non-φ 3-axis control	READY (Controls 1–4, unlaunched)

R2	Q2 bs = 128 baseline control	READY (Phase-9d, ∼3.5 GPU-h)
R3	W2 cross-family re-audit on MAJOR/BROKEN	PARTIAL (intra-family, 8/10 strict CONCORDANT)
AC	#12 category-confusion warning	RESOLVED (§*sec:track-a-dist*, §*sec:track-b-dist*)
AC	#15 auditor model-family caveat	RESOLVED (§*sec:self-grading*)
AC	#16 verdict-on-wrong-testbed audit	PARTIAL (H22 downgrade)

Reproducibility Pointers

- Repository (blind-review redaction will replace this URL): https://github.com/dlmastery/nature_inspired_networks.
- Live dashboard: https://dlmastery.github.io/nature_inspired_networks/.
- Composite-metric fingerprint `d65565e9c7b12d14cbce30a801ecc6753aea3eb148074256bfcc051fa61d0893` (Rule 2; `CompositeFingerprintError` on edit).
- Phase-8 sweep raw seeds: \texttt{experiments/cifar100/<tag>_seed\{0..6\}/}.
- Audit calibration: `audits/AUDIT_CALIBRATION_THIRD_PARTY.md`.
- Phase-9 hill-climb skill: `skills/autoresearch-per-hypothesis-hillclimb/`.